Black-oil simulation of geologic CO_2 sequestration in MRST: theory, verification practice, and a GNU Octave reproduction study

Adriano Cortes[a]

[a] —

Abstract

This report documents the theory and the computational practice behind a full reproduction of the geologic carbon storage (GCS) examples of Saló-Salgado et al. (2024), performed with the MATLAB Reservoir Simulation Toolbox (MRST) under GNU Octave. We review the black-oil formulation used to model CO_2–brine systems, the generation of PVT input data from a thermodynamic mixing model, the Killough–Land model for relative permeability hysteresis, and a Fickian model for molecular diffusion of dissolved CO_2. Each model is exercised on the published example decks: a PVT table verification, the PUNQ-S3 hysteresis validation over 500 years, and a FluidFlower-inspired convective mixing problem at two grid resolutions. We further include a spill-point structural trapping analysis from the `co2lab` module. All results reproduce the published behavior: the fluid object matches the tabulated PVT data to machine precision; hysteresis immobilizes the CO_2 plume by residual trapping and suppresses the mobile gas cap; and grid-level numerical diffusion masks physical diffusion at coarse resolution, as predicted by a simple scaling argument. We close with a portability study that documents the code-level differences between MATLAB and Octave, including a silent grid-corruption defect traced to a malformed block comment, and we discuss its implications for reproducibility of simulation software. An appendix further audits MRST's well-reservoir coupling on the diffusion case's unstructured mesh: we show it approximates the well cell as an axis-aligned Cartesian block rather than using its true PEBI geometry, and quantify the resulting gap against a grid-aware alternative well index. The full reproduction is also released as four executable Jupyter notebooks, one per case, reusing MRST's Octave kernel.

Keywords: geologic carbon storage, black-oil model, MRST, relative permeability hysteresis, molecular diffusion, reproducibility

1. Introduction

Geologic carbon storage (GCS) injects supercritical CO_2 into deep saline aquifers. Four mechanisms retain the injected CO_2: structural and stratigraphic trapping under low-permeability seals, residual trapping as disconnected ganglia after imbibition, solubility trapping by dissolution into brine, and mineral trapping over long times [1]. Numerical simulation is the primary tool to assess these mechanisms at field scale. The MATLAB Reservoir Simulation Toolbox (MRST) [2, 3] provides an open-source platform for such simulations, with full access to the discretization, the nonlinear solvers, and the automatic differentiation (AD) backends.

Saló-Salgado et al. [4] extended the `ad-blackoil` module of MRST with three capabilities that increase the fidelity of black-oil GCS models: (i) generation of CO_2–brine PVT input data from a thermodynamic mixing model; (ii) relative permeability hysteresis of the nonwetting phase, which controls residual trapping; and (iii) molecular diffusion of dissolved CO_2. This report reviews the theory of each extension, describes how MRST implements it, and documents a complete reproduction of the published examples. All runs used GNU Octave 8.4 instead of MATLAB, on the MRST development sources and the example decks published with the paper. The reproduction therefore also serves as a portability and reproducibility study (Section 7).

Section 2 states the governing equations and the discretization. Section 3 presents the spill-point analysis of structural trapping. Sections 4–6 cover the three extensions and their verification cases. Section 7 reports computational aspects and the portability findings. Section 8 concludes. Appendix Appendix A audits the well-index model of Section 6 against a grid-aware alternative for unstructured meshes.

2. Governing equations and discretization

2.1. Black-oil formulation for CO_2–brine systems

The black-oil model describes a multiphase system with pseudo-components whose mutual solubility depends on pressure only. For GCS in saline aquifers, the oleic phase is assigned brine properties, so that the gaseous (CO_2-rich) pseudo-component can dissolve in it [5, 4]. With subscripts o (aqueous/oleic) and g (gaseous), mass conservation of the two pseudo-components reads

$$\frac{\partial}{\partial t}\Big[\phi\big(b_o S_o + R_v b_g S_g\big)\Big] + \nabla\cdot\big(b_o\mathbf{u}_o + R_v b_g \mathbf{u}_g\big) = q_o, \tag{1}$$

$$\frac{\partial}{\partial t}\Big[\phi\big(b_g S_g + R_s b_o S_o\big)\Big] + \nabla\cdot\big(b_g\mathbf{u}_g + R_s b_o \mathbf{u}_o\big) = q_g, \tag{2}$$

where ϕ is porosity, S_α saturation, $b_\alpha = 1/B_\alpha$ the reciprocal formation volume factor, R_s the solution gas–oil ratio (dissolved CO_2 in brine), R_v the vaporized oil–gas ratio (H_2O in the CO_2-rich phase), and q_α source terms. The phase fluxes follow the multiphase extension of Darcy's law,

$$\mathbf{u}_\alpha = -\frac{\mathbf{k}\,k_{r\alpha}(S_\alpha)}{\mu_\alpha}\big(\nabla p_\alpha - \rho_\alpha \mathbf{g}\big), \tag{3}$$

with intrinsic permeability $\mathbf{k}$, relative permeability $k_{r\alpha}$, viscosity μ_α, and phase pressures related by the capillary pressure $p_g - p_o = P_c(S_g)$. The saturations close the system through $S_o + S_g = 1$. Reservoir-condition phase densities follow from the surface densities ρ_α^s:

$$\rho_o = b_o\big(\rho_o^s + R_s\,\rho_g^s\big), \qquad \rho_g = b_g\big(\rho_g^s + R_v\,\rho_o^s\big). \tag{4}$$

2.2. Discretization and solvers in MRST

MRST discretizes (1)–(3) with a cell-centered finite-volume method: two-point flux approximation (TPFA) for the pressure gradient, single-point upstream mobility weighting, and backward Euler in time [2]. For a face ij between cells i and j, the discrete phase flux is

$$v_{\alpha,ij} = \lambda_\alpha^{\mathrm{up}}\, T_{ij}\big(p_{\alpha,i} - p_{\alpha,j} - \rho_{\alpha,ij}\, g\,\Delta z_{ij}\big), \qquad T_{ij} = \Big(T_{i,ij}^{-1} + T_{j,ij}^{-1}\Big)^{-1}, \tag{5}$$

where $T_{i,ij}$ are half-face transmissibilities built from the cell permeability and geometry, and $\lambda_\alpha^{\mathrm{up}} = (k_{r\alpha}/\mu_\alpha)^{\mathrm{up}}$ is evaluated in the upstream cell. The fully implicit residual system is linearized by Newton's method. MRST assembles the Jacobians by automatic differentiation and organizes secondary variables (densities, mobilities, relative permeabilities, component fluxes) as *state functions* attached to the model [3]. This design is what allows the hysteresis and diffusion extensions of Sections 5–6 to replace one state function each, without changes to the assembly loop.

In this work all systems were solved with a sparse direct solver (`BackslashSolverAD`), a Newton line search, and an iteration-targeting time-step controller. The Octave environment provides no compiled (MEX) acceleration; Section 7 quantifies the consequences.

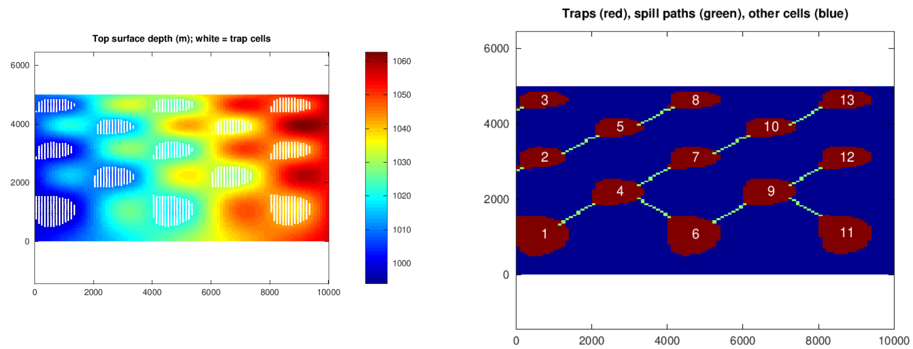

Figure 1: Spill-point analysis of the synthetic sandbox aquifer. Left: top-surface depth (m); white dots mark cells inside structural traps. Right: the 13 numbered traps (red), the spill paths connecting them (green), and the remaining cells (blue). CO_2 that leaks from a trap migrates along the spill path to the next trap up-dip.

3. Structural trapping: spill-point analysis

Before dynamic simulation, geometric analysis of the caprock surface already quantifies the structural trapping potential. The `co2lab` module builds a top-surface grid Γ from the 3D grid and analyzes its depth function $z(\mathbf{x})$ [6]. A *trap* is the interior of a closed depth contour around a local minimum of z (a dome apex); its *spill point* is the saddle where CO_2 overflows to the next trap. The traps and the connecting *spill paths* form a directed tree that predicts the long-term migration of buoyant CO_2 without solving any flow equation.

We exercised this analysis on the introductory example of `co2lab-spillpoint`: a synthetic 10×5 km, 50 m thick sandbox at 1 km depth, with a dipping, sinusoidally perturbed top surface (100×100 top-surface cells, porosity 0.25). The edge-based algorithm of `trapAnalysis` identified 13 traps with a total capacity of $5.099 \times 10^6\,\mathrm{m}^3$, which is 40.8% of the total pore volume ($1.25 \times 10^7\,\mathrm{m}^3$); the five largest traps hold 55.2% of that capacity. Figure 1 shows the depth map with the trap cells and the trap–spill-path network.

4. PVT data generation for CO_2–brine black-oil models

4.1. Thermodynamic model

Black-oil simulators consume tabulated PVT data. For CO_2–brine systems, the tables must encode the mutual solubility of CO_2 and H_2O and the resulting density and viscosity changes. The implementation of [4] follows Hassanzadeh et al. [5]: phase compositions come from the fugacity–activity equilibrium model of Spycher et al. [7, 8] combined with the activity coefficients of Duan and Sun [9], which admits NaCl, KCl, $CaCl_2$, and seawater brines; the model is valid to approximately 100 °C and 600 bar. Gas-phase molar volumes follow the Redlich–Kwong equation of state [10]. Brine density follows Rowe and Chou [11], extended in p–T range by Batzle and Wang [12], with the density increase due to dissolved CO_2 from Garcia's apparent molar volume correlation [13]. Viscosities use Islam and Carlson [14] for the aqueous phase and Fenghour et al. [15] for CO_2, with Davidson's mixing rule [16] when the gas phase contains vaporized H_2O.

The black-oil quantities follow from the equilibrium compositions. The solution gas–oil ratio R_s is the volume of dissolved CO_2, measured at surface conditions, per surface volume of brine, and B_o is the ratio of the reservoir-condition aqueous volume to its surface volume. If vaporized water is included, the gas-phase analogues read [4]

$$R_v = \frac{V^s_{v\mathrm{H_2O}}}{V^s_{\mathrm{CO_2}}} = \frac{y_{\mathrm{H_2O}}\,\bar{\rho}^s_{\mathrm{CO_2}}}{\bar{\rho}^s_{\mathrm{H_2O}}\,y_{\mathrm{CO_2}}}, \qquad B_g = \frac{V^r_{\mathrm{g}}}{V^s_{\mathrm{CO_2}}} = \frac{\bar{\rho}^s_{\mathrm{CO_2}}}{\bar{\rho}^r_{\mathrm{g}}\,(1 - \upsilon_{\mathrm{H_2O}})}, \qquad (6)$$

where y and υ are mole and mass fractions in the gaseous phase and $\bar{\rho}$ denotes molar density. The function `pvtBrineWithCO2BlackOil` evaluates this chain over a requested p–T–salinity range and writes ECLIPSE-format `PVTO` (live oil, i.e., brine with dissolved CO_2) and `PVDG` (dry gas) tables.

4.2. Verification

The published deck `example_co2brine_3reg_unsatVals.DATA` embeds tables generated at $T = 80\,^\circ\mathrm{C}$, salinity 10^5 ppm, and $p \in [1, 600]$ bar. Reading this deck with `initDeckADIFluid` produces the MRST fluid object; evaluating its b_g, μ_g, R_s^{sat}, b_o, μ_o functions and reconstructing densities through (4) must return the tabulated values. Figure 2 shows the comparison; the maximum relative difference over all four quantities is 2.2×10^{-16}, i.e., machine precision, which verifies the deck round-trip (table $\rightarrow$ deck $\rightarrow$ interpolated fluid object). The curves also show the physics of interest for GCS: the supercritical density transition of CO_2 (223, 734, and 933 kg/m^3 at 100, 300, and 600 bar) and the density increase of brine upon CO_2 dissolution, which drives the convective mixing of Section 6.

5. Relative permeability hysteresis

5.1. Killough's model with Land trapping

Relative permeability depends on the saturation path, not only on the current saturation: pore-scale contact-angle hysteresis and nonwetting phase

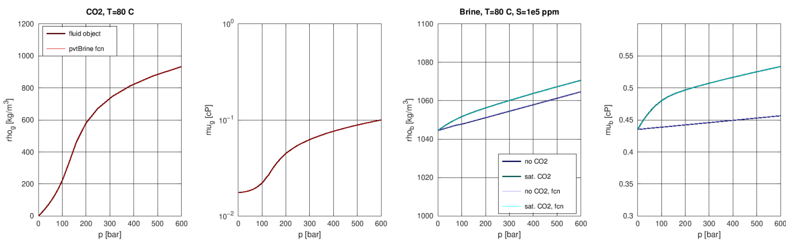

Figure 2: Verification of the PVT deck round-trip at $T = 80\,^\circ\mathrm{C}$, $S = 10^5$ ppm. Thick lines: MRST fluid object; thin lines: tabulated output of `pvtBrineWithCO2BlackOil`. From left to right: CO_2 density, CO_2 viscosity (log scale), brine density without and with dissolved CO_2, brine viscosity. The curves coincide to machine precision; compare with Fig. 2 of [4].

disconnection make the imbibition curve differ from the drainage curve [17]. In GCS, CO_2 is the nonwetting phase in most aquifers; during post-injection migration, brine re-imbibes at the trailing edge of the plume, disconnects CO_2 ganglia, and immobilizes them (residual trapping). A drainage-only model misses this mechanism and overpredicts the mobile gas cap.

Killough's model [18] computes the scanning curve between the primary drainage curve k_{rg}^d and the bounding imbibition curve k_{rg}^{ib}. While the current saturation S_g lies below the historical maximum S_{gi}, the scanning relative permeability is

$$k_{rg}^i(S_g) = k_{rg}^{ib}(S_g^*)\,\frac{k_{rg}^d(S_{gi})}{k_{rg}^d(S_{g,\max})}, \qquad S_g^* = S_{gt,\max} + \frac{(S_g - S_{gt})(S_{g,\max} - S_{gt,\max})}{S_{gi} - S_{gt}}, \tag{7}$$

where $S_{g,\max}$ is the maximum attainable gas saturation and $S_{gt,\max}$ the corresponding maximum trapped saturation. The trapped gas saturation for the current cycle follows Land's model [19],

$$S_{gt} = S_{g,\min} + \frac{S_{gi} - S_{g,\min}}{1 + C\,(S_{gi} - S_{g,\min})}, \qquad C = \frac{1}{S_{gt,\max} - S_{g,\min}} - \frac{1}{S_{g,\max} - S_{g,\min}}, \tag{8}$$

with $S_{g,\min} = 0$ on primary drainage. For numerical robustness the implementation replaces the denominator $1+C(S_{gi}-S_{g,\min})$ by $A+C(S_{gi}-S_{g,\min})$ with $A = 1 + a(S_{g,\max} - S_{gi})$ and default $a = 0.1$, as ECLIPSE does [4]. MRST reads the hysteresis options from the `EHYSTR` keyword and the imbibition saturation tables from the deck, and evaluates (7) inside a dedicated state function, `HystereticRelativePermeability`, that replaces the default relative permeability of the validated model. Figure 3 shows the three

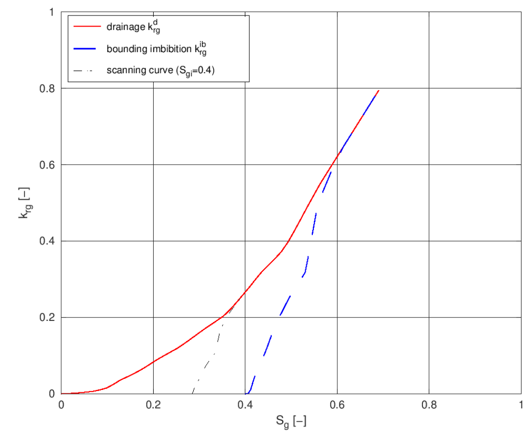

Figure 3: Hysteretic gas relative permeability of the PUNQ-S3 deck: primary drainage curve, bounding imbibition curve, and the Killough scanning curve for a flow reversal at $S_{gi} = 0.4$. The scanning curve reaches zero at the Land trapped saturation $S_{gt}(S_{gi}) \approx 0.29$. Compare with Fig. 3 of [4].

curves for the PUNQ-S3 rock (Berea sandstone data of Oak, used by [17]), computed from the published deck.

5.2. Validation on the PUNQ-S3 model

Juanes et al. [17] adapted the PUNQ-S3 geologic model into a GCS benchmark: a $19 \times 28 \times 5$ corner-point grid (1,761 active cells) of an anticline aquifer with heterogeneous permeability (0.5–1000 mD), eight injectors at $18\,\mathrm{rm}^3/\mathrm{day}$ each for 10 years, 500 years of total simulation, immiscible water–gas physics, and pore volumes multiplied by 10^3 on the boundary ring to mimic an open aquifer. We reproduced both published cases from the same deck (`CASE2.DATA`): Case 1 deactivates hysteresis; Case 2 activates Killough hysteresis in the gas phase.

Figure 4 shows the CO_2 saturation in all five layers at $t = 500$ y, and Fig. 5 the plume metrics in time. The two cases coincide until injection stops at $t = 10$ y, because hysteresis only acts on saturation reversals. Afterwards they diverge: without hysteresis the plume migrates up-dip, drains the invaded region, and collects into a compact, high-saturation gas cap (49 cells with $S_g > 0.05$, mean plume saturation 0.505 at 500 y); with hysteresis the trailing edge traps CO_2 residually along the migration path, the invaded area freezes at its maximum extent (138 cells), and the mean plume saturation drops to 0.192. Table 1 summarizes the metrics. This is precisely the

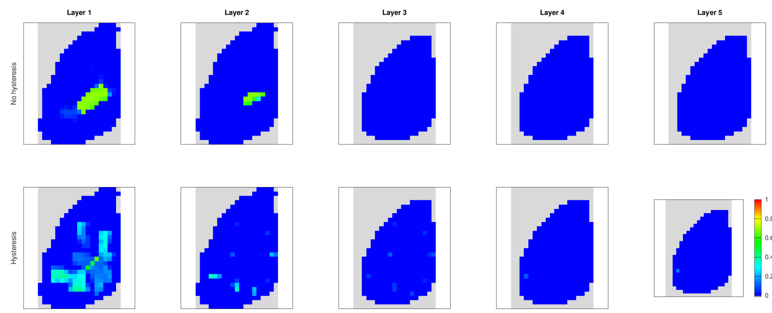

Figure 4: PUNQ-S3: CO_2 saturation per layer (left to right: layers 1–5, top of the formation to its base) at $t = 500$ y. Top row: no hysteresis; bottom row: Killough hysteresis. Gray marks inactive cells. Hysteresis suppresses the compact gas cap and leaves a broad, low-saturation, residually trapped plume. Compare with Fig. 4 of [4].

Table 1: PUNQ-S3 plume metrics at $t = 500$ y.

Metric	No hysteresis	Hysteresis
Maximum S_g	0.690	0.641
Cells with $S_g > 0.05$	49	138
Cells with $S_g > 0.3$	36	23
Mean S_g where $S_g > 0.05$	0.505	0.192

behavior reported in Fig. 4 of [4] and in [17], where the same contrast was validated against ECLIPSE 100.

6. Molecular diffusion of dissolved CO_2

6.1. Model and numerical diffusion

When CO_2 dissolves into brine, the mixture is denser than the resident brine and sinks in Rayleigh–Taylor fingers, which accelerates dissolution trapping. Concentration gradients also drive a diffusive mass flux. The total macroscale flux of component γ in phase α decomposes as [20, 4]

$$\mathbf{J}_\alpha^\gamma = \underbrace{\rho_\alpha\,\chi_\alpha^\gamma\,\mathbf{u}_\alpha}_{\text{advective}} - \underbrace{\phi\,\rho_\alpha S_\alpha\,D_\alpha^\gamma\,\nabla\chi_\alpha^\gamma}_{\text{diffusive (Fickian)}}, \tag{9}$$

where χ_α^γ is the mass fraction and D_α^γ a scalar pseudo-diffusivity that lumps molecular diffusivity and tortuosity; for CO_2 in brine the molecular value is

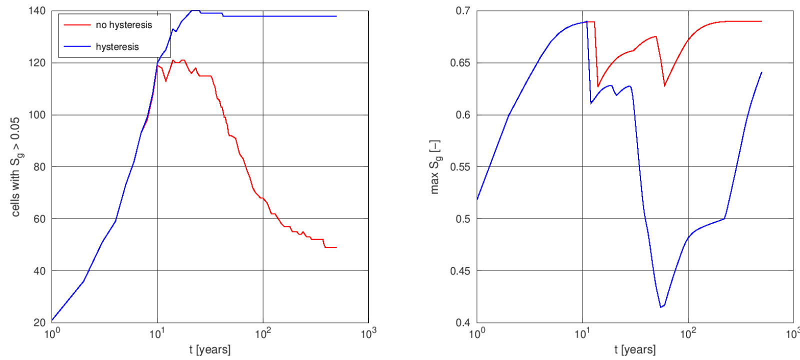

Figure 5: PUNQ-S3 plume evolution. Left: number of cells with $S_g > 0.05$; right: maximum gas saturation. The curves coincide during injection ($t \leq 10$ y) and diverge after flow reversal, when the hysteretic scanning curves activate.

$\mathcal{D} \sim 2$–5×10^{-9} m^2/s. MRST implements (9) by replacing the `ComponentTotalFlux` state function of `GenericBlackOilModel` with `CO2TotalFluxWithDiffusion`, which adds the second term with a TPFA-style face diffusivity.

Whether the diffusive term matters numerically depends on the grid. A first-order upstream scheme carries a numerical diffusion $D_{\text{num}} \sim u\,h$, with u the Darcy velocity and h the cell size. For a laboratory-scale problem with $k = 100$ mD, $\mu = 0.8$ cP, and a pressure gradient of 10 mbar/m, $u \approx 1.2 \times 10^{-7}$ m/s, so $h = 1$ cm gives $D_{\text{num}} \sim 10^{-9}$ m^2/s—larger than the physical diffusivity—while $h = 2.5$ mm brings the two within reach of each other. Physical diffusion therefore only shapes the solution when the grid is fine enough, and conversely, coarse-grid simulations already contain more (numerical) diffusion than the physics requires [4].

6.2. A FluidFlower-inspired convective mixing problem

The test case is a quasi-2D stratigraphic section (1×0.66 m, one 1-cm-thick cell layer) inspired by the FluidFlower benchmark rig [21], repositioned at 1 km depth so that CO_2 is supercritical. The PEBI mesh conforms to a fault structure (fault permeability 10 mD inside a 1 D reservoir); sealing units are removed from the flow domain, and lateral boundary cells receive a 10^5 pore-volume multiplier to emulate an open aquifer. CO_2 is injected in the lower-right reservoir at 8 mL/min (surface conditions) for one day, with rate ramps, and the model then runs to $t = 30$ days with dissolution active (`disgas`).

We ran the case on two meshes: coarse ($h \approx 1$ cm, 8,206 cells) and

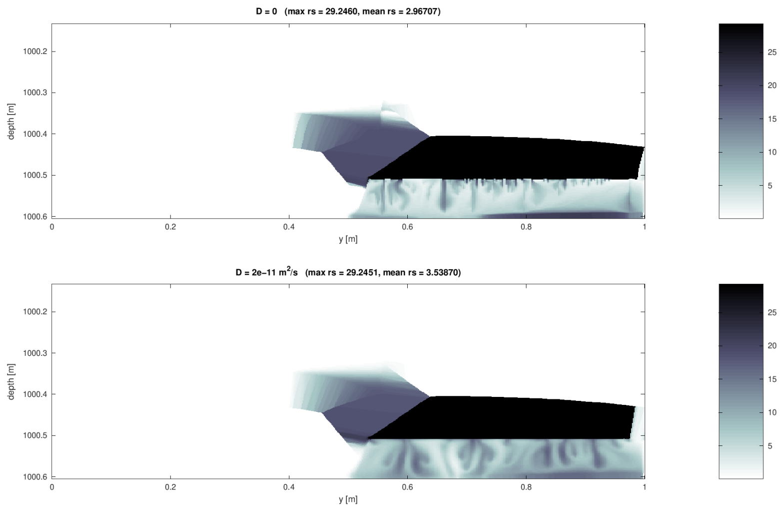

Figure 6: Fine mesh ($h \approx 2.5$ mm, 105,279 cells): dissolved CO_2 (R_s) at $t = 30$ d without (top) and with (bottom) molecular diffusion ($D = 2 \times 10^{-11}$ m^2/s). Finger width and spacing increase with the diffusivity. Compare with Fig. 6(a,b) of [4].

fine ($h \approx 2.5$ mm, 105,279 cells, the resolution of Fig. 6 in [4]), each with $D = 0$ and $D = 2 \times 10^{-11}$ m^2/s. Figure 7 shows the dissolved CO_2 field (R_s) at $t = 30$ d on the coarse mesh: a CO_2-saturated gas accumulation under the sealing horizon, and sinking CO_2-rich brine below it. At this resolution the two diffusivities give nearly identical fields (mean R_s 4.06 vs. 4.13 Sm^3/Sm^3; plume cell count 1,027 vs. 1,105), which confirms the scaling argument: numerical diffusion at $h = 1$ cm dominates the physical value. Figure 6 repeats the comparison on the fine mesh, where the convective fingers are resolved and the physical diffusivity becomes visible, as in Fig. 6 of [4]: without diffusion the fingers below the gas accumulation are thin and closely spaced; with $D = 2 \times 10^{-11}$ m^2/s they are wider, smoother, and farther apart. Diffusion also enhances dissolution trapping: the mean dissolved ratio rises from 2.97 to 3.54 Sm^3/Sm^3, and the invaded region grows from 9,574 to 10,389 cells.

7. Computational practice and portability

7.1. Runtimes

All simulations ran fully implicitly with a sparse direct linear solver, Newton line search, and iteration-targeting time steps, in GNU Octave 8.4

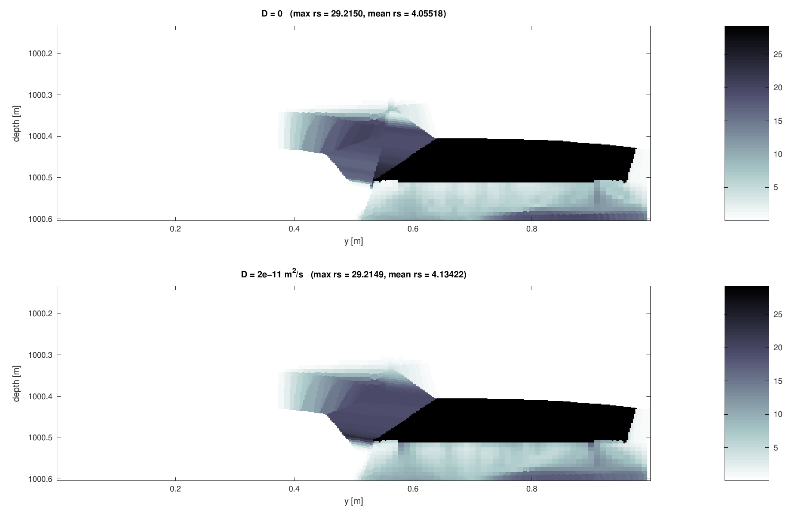

Figure 7: Coarse mesh ($h \approx 1$ cm, 8,206 cells): dissolved CO_2 (R_s, $\mathrm{Sm}^3/\mathrm{Sm}^3$) at $t = 30$ d, without (top) and with (bottom) molecular diffusion. Dark: CO_2-saturated brine under the gas accumulation; the halo below is the sinking CO_2-rich brine. At this resolution numerical diffusion masks the physical diffusivity.

on a Linux workstation, single-threaded, without any MEX acceleration. Table 2 lists the runtimes. For reference, [4] reports 6857 s for the 90,000-cell Johansen field case in MATLAB with a direct solver, reduced to 417 s with a compiled AD backend and an algebraic multigrid CPR solver; that case was out of scope here for exactly this reason.

7.2. MATLAB-to-Octave portability

The reproduction required a set of well-defined substitutions, all outside the numerical kernels: the `table` class (used only by the PVT table writer; the shipped tables were used instead), `deval` (replaced by explicit midpoint integration of the hydrostatic ODE), `pdist2` (replaced by a direct norm), `griddedInterpolant` (MRST ships an Octave replacement of `fastInterpTable` that must shadow the MATLAB version), and `delaunayTriangulation` (a minimal shim exposing only the `edges` method suffices for the PEBI grid generator).

One finding is of general interest for simulation-software reproducibility. The PEBI meshes of Section 6 initially came out with 38% of cells having negative volume, without any error. The cause was a malformed block comment in `sortEdges.m` of the `upr` module: the `%}` closer was glued to the

Table 2: Reproduction runtimes (GNU Octave 8.4, single thread, direct linear solver).

Case	Cells	Report steps	Runtime [s]
PUNQ-S3, no hysteresis	1,761	100	33.5
PUNQ-S3, hysteresis	1,761	100	51.4
Diffusion, coarse, D=0	8,206	69	167.9
Diffusion, coarse, D=2×10^{-11}	8,206	69	166.9
Diffusion, fine, D=0	105,279	69	3531.7
Diffusion, fine, D=2×10^{-11}	105,279	69	2751.0

end of the copyright line. MATLAB tolerates this; Octave parses the block as unterminated, silently treats the whole function body as a comment, and the function degenerates to an identity map. The unsorted cell topology then produced inverted prisms during extrusion—three calls downstream of the defect. A second file (`mrstGridWithFullMappings.m`) lost its entire function definition to the same pattern. The defect exists only in MRST's subtree copy (the upstream UPR repository is well-formed), which points to a mass copyright-update script as the origin. The fix was submitted upstream.[1] The episode illustrates two practical lessons: (i) cross-interpreter parsing differences can corrupt *results* rather than raise errors, so geometric invariants (here: positive cell volumes and face areas) are worth asserting after every grid-construction step; and (ii) automated rewriting of source files, however trivial, deserves a syntax check in all supported interpreters.

7.3. Companion Jupyter notebooks

The reproduction is also released as four executable Jupyter notebooks (`reproductions/salo2024-gcs/notebooks/`), one per case (structural trapping, PVT, hysteresis, diffusion), running MRST through an Octave Jupyter kernel with markdown narrative paraphrased from this report. The expensive cases (hysteresis, diffusion) default to loading the cached `results/*.mat` states rather than re-simulating, with a boolean flag to opt into a live re-run; both paths were verified to reproduce the numbers in this report exactly. Building them surfaced one further portability pitfall, specific to the interactive kernel rather than to headless scripts: this Octave build has no `qt` graphics toolkit, and `fltk` is needed for the 3-D patch plots (`plotGrid` on a 3-D grid) that gnuplot cannot draw; but forcing `fltk` as the session-wide

[1] https://github.com/SINTEF-AppliedCompSci/MRST/pull/37

default silently renders every `imagesc` plot – the type used throughout this report's own figures – as a blank raster, with no error. The fix is to keep gnuplot as the default and switch to `fltk` only in the specific cell that draws a 3-D patch plot.

8. Conclusions

We reproduced, in GNU Octave, the three `ad-blackoil` extensions for geologic CO_2 storage published by Saló-Salgado et al. [4], plus a spill-point structural trapping analysis. The PVT deck round-trip is exact to machine precision. The PUNQ-S3 validation reproduces the published effect of relative permeability hysteresis: residual trapping freezes the plume at its maximum extent and suppresses the mobile gas cap, with divergence starting exactly at flow reversal. The convective mixing example reproduces both the physics (dissolution-driven density fingering) and the numerical analysis: the physical diffusivity only matters once the grid-level numerical diffusion drops below it. The full reproduction, including a 105k-cell fine-mesh run, is feasible on open-source tooling in minutes to tens of minutes per case; the exercise also uncovered and fixed a silent grid-corruption defect, which underlines the value of independent reproduction across interpreters. Appendix Appendix A shows that MRST's well-index model on this PEBI mesh is a Cartesian bounding-box approximation of Peaceman's formula, not a grid-aware one: the gap against a Voronoi-specific alternative grows with cell irregularity, though it stays within this problem's large injectivity margin.

Appendix A. Peaceman vs. Palagi–Aziz well index on a PEBI mesh

Section 2 characterized MRST's well-reservoir coupling as Peaceman-type [22]. This appendix examines that characterization on the diffusion case's own mesh (Section 6), whose injector cell is a PEBI polygon, not a Cartesian block, and quantifies the effect of a grid-aware alternative.

Appendix A.1. The coupling and well index used by MRST

For a single perforated cell, the rate/pressure relation and well index implemented in `core/utils/computeWellIndex.m` are

$$q = -WI \cdot \mathrm{mob} \cdot \left(p_{\mathrm{cell}} - p_{bh} - \Delta p_{\mathrm{hydrostatic}}\right), \qquad WI = \frac{2\pi\sqrt{k_x k_y}\, h}{\ln(r_0/r_w) + \mathrm{skin}}, \tag{A.1}$$

with the anisotropic equivalent radius of Peaceman [23]

$$r_0 = \frac{0.28\sqrt{\sqrt{k_y/k_x}\,\Delta x^2 + \sqrt{k_x/k_y}\,\Delta y^2}}{(k_y/k_x)^{1/4} + (k_x/k_y)^{1/4}}. \tag{A.2}$$

Equation (A.2) assumes the well cell *is* a rectangle of known $\Delta x, \Delta y$ – true by construction on a Cartesian or corner-point grid, but undefined for an irregular PEBI polygon.

Appendix A.2. The bounding-box approximation on a PEBI grid

For any grid with a node list, MRST's `cellDims.m` supplies $\Delta x, \Delta y$ to `computeWellIndex` as the axis-aligned bounding box of the well cell's own nodes, whatever its true shape: it takes the minimum and maximum node coordinates and sets $\Delta x = M_x - m_x$, $\Delta y = M_y - m_y$. This never inspects the well cell's actual neighbors, connection transmissibilities, or number of sides – two very differently shaped cells with the same bounding box receive the same well index.

Appendix A.3. Palagi and Aziz's well index for Voronoi grids

Palagi and Aziz [24, 25] derive an equivalent radius directly from the well cell's actual connections, so it is defined for any polygon. Assuming radial flow around the well-cell gridpoint out to each neighbor j,

$$p_j - p_w = \frac{qB\mu\,\ln(L_{ij}/r_w)}{\theta kh}, \tag{A.3}$$

and requiring this to be consistent with the discretized single-phase flow equation for the well cell,

$$q = \sum_j \frac{T_{ij}}{B\mu}(p_j - p_o) = \sum_j \frac{T_{ij}}{B\mu}(p_j - p_w) - \sum_j \frac{T_{ij}}{B\mu}(p_o - p_w), \tag{A.4}$$

gives the equivalent radius as a transmissibility-weighted average over the real connections,

$$\ln r_{eq} = \frac{\sum_j T_{ij}\ln L_{ij} - \theta kh}{\sum_j T_{ij}}, \qquad WI = \frac{\theta kh}{\ln(r_{eq}/r_w) + \text{skin}}, \tag{A.5}$$

where T_{ij} is the two-point transmissibility of connection ij, L_{ij} the distance between cell centroids, and θ the angle open to flow ($\theta = 2\pi$ for a fully interior well cell, matching the implicit 2π in Eq. (A.1)), which keeps the

Table A.3: Well index at $r_w = 1$ mm, skin = 0: Peaceman (MRST default) vs. Palagi–Aziz (Eq. A.5).

Location	WI Peaceman	WI Palagi–Aziz	Difference
Reservoir well (original)	9.034×10^{-14}	8.157×10^{-14}	-9.71%
Elongated fault-strip cell	5.555×10^{-16}	4.583×10^{-16}	-17.50%

two models directly comparable). Equation (A.5) reduces to Peaceman's formula for a square Cartesian cell but stays correct for an irregular polygon with any number of neighbors. We implemented Eq. (A.5) as a standalone function (`scripts/computeWellIndexPalagiAziz.m`) that reuses MRST's own `cellDims` and `computeTrans`, matching kh, skin, and r_w exactly with Eq. (A.1) so the two well indices differ only in how r_{eq} is derived; `addWell` accepts a precomputed `WI` directly, so no MRST internals needed modifying.

Appendix A.4. Comparison

Both models were evaluated at $r_w = 1$ mm, skin = 0, on the coarse mesh of Section 6, at two locations: the paper's actual injection cell (in the reservoir matrix, nearly a cube, $\Delta x \approx \Delta y \approx \Delta z \approx 1$ cm), and the most elongated fault-strip cell reachable near that same location (bounding-box aspect ratio $R_y \approx 2.4$, found by ranking all fault cells with `scripts/find_fault_cell.m`) – Palagi and Aziz [25] flag $R_y > 2$ as exactly the regime where the Cartesian formula stops being reliable. Table A.3 shows the well-index gap growing with cell irregularity, as expected.

Because this well is rate-controlled, WI only sets the drawdown $p_{\text{cell}} - p_{bh}$ needed to deliver the fixed 8 mL/min – it never changes how much CO_2 enters the reservoir, so the saturation and dissolved-CO_2 fields should be governed by rate, not by WI, and only the bottom-hole pressure should respond. We reran the coarse-mesh, $D = 0$ case with each well index at both locations (Table A.4): the field-wide difference stays at the level of solver noise at both locations, but the bottom-hole-pressure gap grows about 230× from the reservoir well to the fault-strip cell, tracking the larger well-index gap and the fault's lower permeability (10 mD vs. 1000 mD, i.e. less injectivity headroom).

Appendix A.5. Discussion

MRST's well model is Peaceman-type, but the equivalent radius it computes for a PEBI cell is a bounding-box stand-in for the Cartesian formula,

Table A.4: Effect of the well-index model on the coarse-mesh, $D = 0$ simulation (max. absolute difference over the full 30-day schedule).

Location	Max $\lvert\Delta p_{bh}\rvert$ [Pa]	Max $\lvert\Delta r_s\rvert$	Max $\lvert\Delta S_g\rvert$
Reservoir well	0.35	1.6×10^{-6}	3.8×10^{-9}
Elongated fault-strip cell	80.0	6.5×10^{-6}	1.5×10^{-7}

not a generalization of it. That gap is real and grows with cell irregularity, here about twice as large at an elongated fault-strip cell as at a well-conditioned reservoir cell. Whether it matters for a given simulation is a separate question: for this rate-controlled injector it changes only the computed bottom-hole pressure, never the injected mass, and even the largest pressure gap found (80 Pa against a $\sim$9.9 MPa reservoir pressure) is negligible at this problem's flow rate – the reservoir has far more injectivity headroom than this well ever calls on. A pressure-controlled well, a higher rate, or a lower-permeability target would be expected to show a larger effect, since drawdown itself would no longer be negligible.

Acknowledgments

This report and the reproduction runs were prepared with the assistance of Claude Code (Anthropic). MRST is developed by the Applied Computational Science group at SINTEF Digital. The example decks and code are from L. Saló-Salgado's `mrst-adblackoil-gcs` repository and MRST's `co2lab-mit` module.

References

[1] S. Benson, P. Cook, et al., Underground geological storage, in: IPCC Special Report on Carbon Dioxide Capture and Storage, Cambridge University Press, 2005, pp. 196–276.

[2] K.-A. Lie, An Introduction to Reservoir Simulation Using MATLAB/GNU Octave: User Guide for the MATLAB Reservoir Simulation Toolbox (MRST), Cambridge University Press, 2019.

[3] K.-A. Lie, O. Møyner (Eds.), Advanced Modelling with the MATLAB Reservoir Simulation Toolbox, Cambridge University Press, 2021.

[4] L. Saló-Salgado, O. Møyner, K.-A. Lie, R. Juanes, Three-dimensional simulation of geologic carbon dioxide sequestration using MRST, Advances in Geo-Energy Research 14 (2024) 34–48. https://doi.org/10.46690/ager.2024.10.06

[5] H. Hassanzadeh, M. Pooladi-Darvish, A. M. Elsharkawy, D. W. Keith, Y. Leonenko, Predicting PVT data for CO_2–brine mixtures for black-oil simulation of CO_2 geological storage, International Journal of Greenhouse Gas Control 2 (2008) 65–77.

[6] H. M. Nilsen, K.-A. Lie, O. Andersen, Analysis of CO_2 trapping capacities and long-term migration for geological formations in the Norwegian North Sea using MRST-co2lab, Computers & Geosciences 79 (2015) 15–26.

[7] N. Spycher, K. Pruess, J. Ennis-King, CO_2–H_2O mixtures in the geological sequestration of CO_2. I. Assessment and calculation of mutual solubilities from 12 to 100°C and up to 600 bar, Geochimica et Cosmochimica Acta 67 (2003) 3015–3031.

[8] N. Spycher, K. Pruess, CO_2–H_2O mixtures in the geological sequestration of CO_2. II. Partitioning in chloride brines at 12–100°C and up to 600 bar, Geochimica et Cosmochimica Acta 69 (2005) 3309–3320.

[9] Z. Duan, R. Sun, An improved model calculating CO_2 solubility in pure water and aqueous NaCl solutions from 273 to 533 K and from 0 to 2000 bar, Chemical Geology 193 (2003) 257–271.

[10] O. Redlich, J. N. S. Kwong, On the thermodynamics of solutions. V. An equation of state. Fugacities of gaseous solutions, Chemical Reviews 44 (1949) 233–244.

[11] A. M. Rowe Jr., J. C. S. Chou, Pressure–volume– temperature–concentration relation of aqueous NaCl solutions, Journal of Chemical and Engineering Data 15 (1970) 61–66.

[12] M. Batzle, Z. Wang, Seismic properties of pore fluids, Geophysics 57 (1992) 1396–1408.

[13] J. E. Garcia, Density of aqueous solutions of CO_2, Report LBNL-49023, Lawrence Berkeley National Laboratory, 2001.

[14] A. W. Islam, E. S. Carlson, Viscosity models and effects of dissolved CO_2, Energy & Fuels 26 (2012) 5330–5336.

[15] A. Fenghour, W. A. Wakeham, V. Vesovic, The viscosity of carbon dioxide, Journal of Physical and Chemical Reference Data 27 (1998) 31–44.

[16] T. A. Davidson, A simple and accurate method for calculating viscosity of gaseous mixtures, Report of Investigations 9456, US Bureau of Mines, 1993.

[17] R. Juanes, E. J. Spiteri, F. M. Orr Jr., M. J. Blunt, Impact of relative permeability hysteresis on geological CO_2 storage, Water Resources Research 42 (2006) W12418.

[18] J. E. Killough, Reservoir simulation with history-dependent saturation functions, SPE Journal 16 (1976) 37–48.

[19] C. S. Land, Calculation of imbibition relative permeability for two- and three-phase flow from rock properties, SPE Journal 8 (1968) 149–156.

[20] J. Bear, Dynamics of Fluids in Porous Media, Dover, New York, 1972.

[21] B. Flemisch, et al., The FluidFlower validation benchmark study for the storage of CO_2, Transport in Porous Media 151 (2024) 865–912.

[22] D. W. Peaceman, Interpretation of well-block pressures in numerical reservoir simulation, SPE Journal 18 (1978) 183–194.

[23] D. W. Peaceman, Interpretation of well-block pressures in numerical reservoir simulation with nonsquare grid blocks and anisotropic permeability, SPE Journal 23 (1983) 531–543.

[24] C. L. Palagi, K. Aziz, Use of Voronoi grid in reservoir simulation, SPE Advanced Technology Series 2 (2) (1994) 69–77.

[25] C. L. Palagi, K. Aziz, Modeling vertical and horizontal wells with Voronoi grid, SPE Reservoir Engineering 9 (1) (1994) 15–21.